ASSIGNMENT-2

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BATCH-16

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Task 1: Book Class Generation (Using Cursor AI)

Python Code: Book Class

Thonny - <untitled> @ 5 : 29

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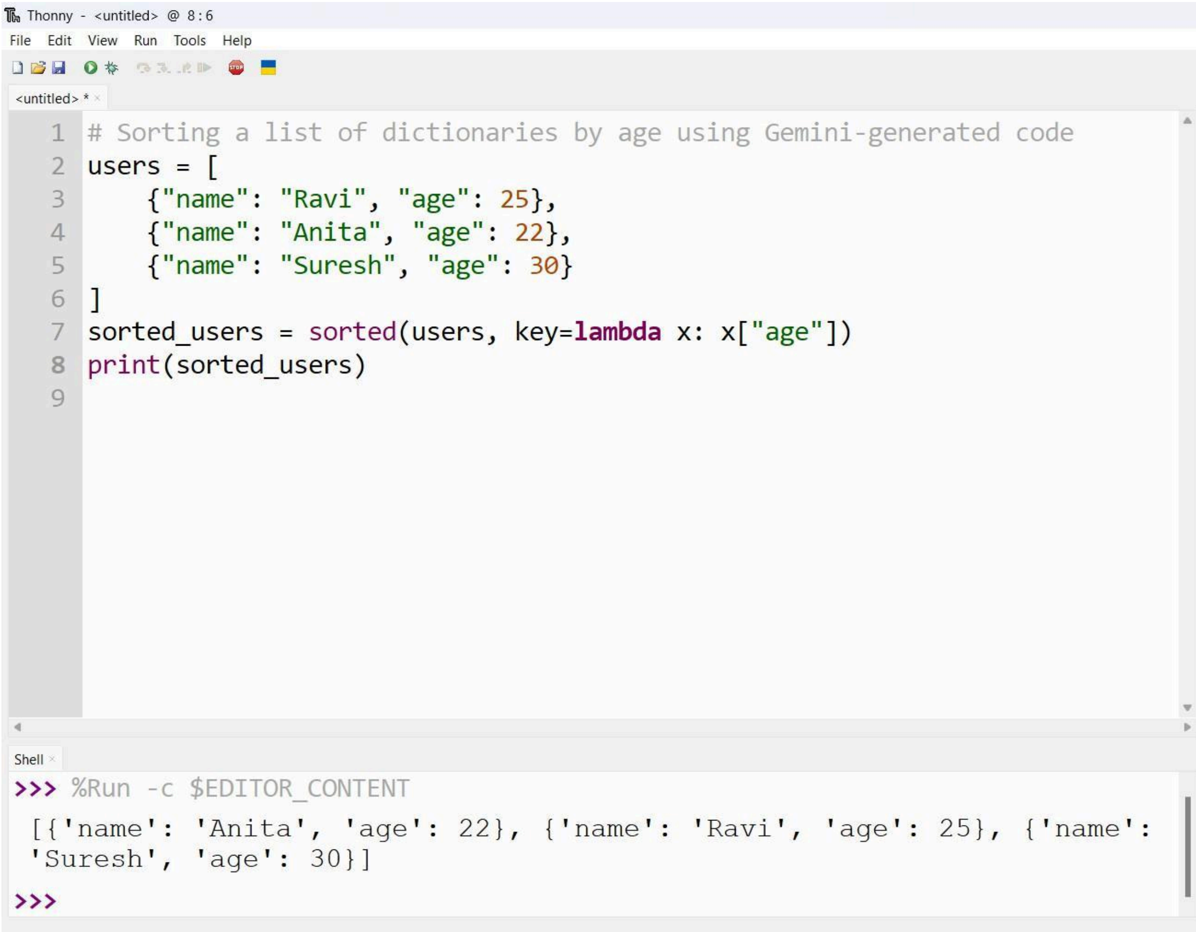
```python
# Book class representing a simple library book
class Book:
    def __init__(self, title, author):
        self.title = title
        self.author = author
    def summary(self):
        return f"Title: {self.title}, Author: {self.author}"

# Example usage
book1 = Book("Artificial Intelligence", "Stuart Russell")
print(book1.summary())
```

Shell

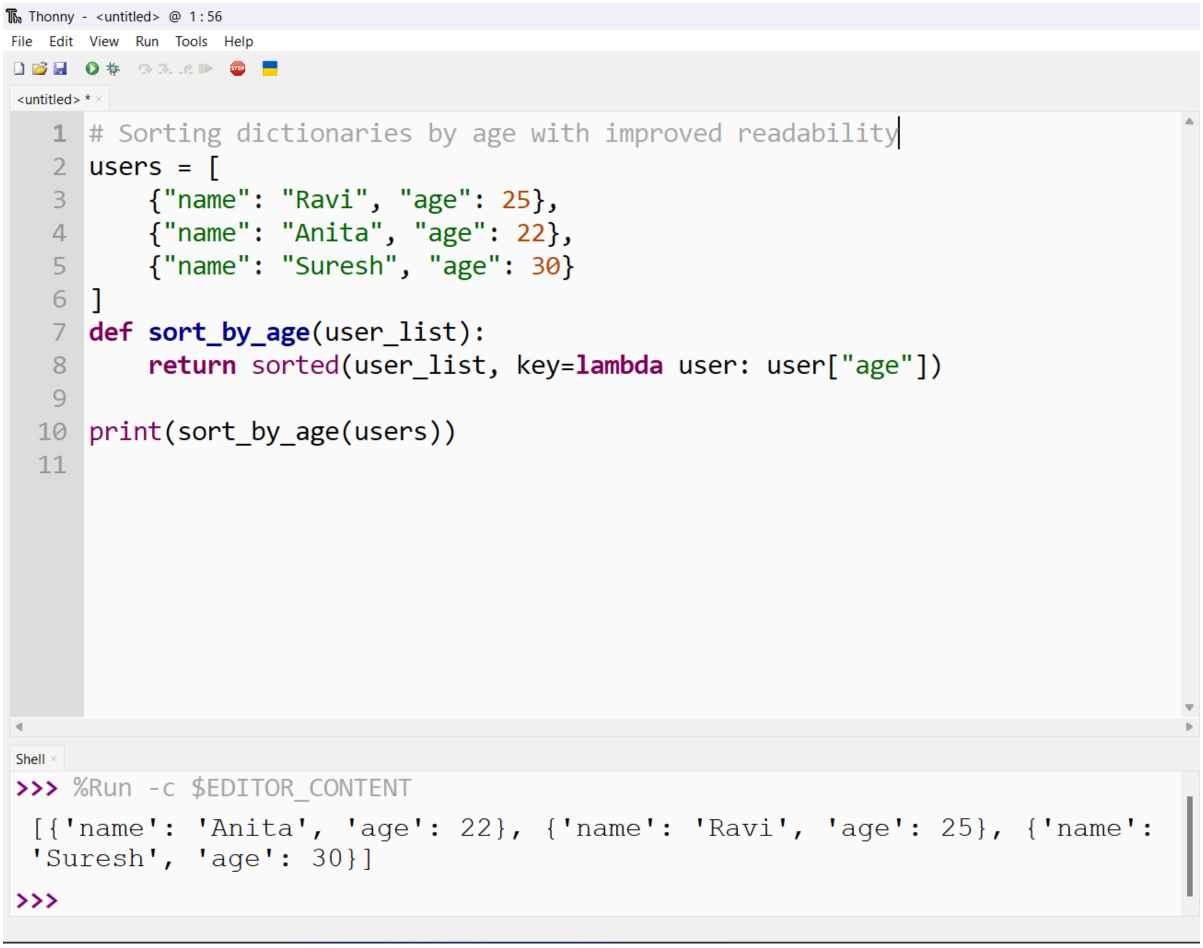
```
>>> %Run -c $EDITOR_CONTENT
 Title: Artificial Intelligence, Author: Stuart Russell
>>>
```

Task 2: Sorting Dictionaries with AI

◆ **Using Google Gemini**



Using Cursor AI




```python
# Sorting dictionaries by age with improved readability
users = [
    {"name": "Ravi", "age": 25},
    {"name": "Anita", "age": 22},
    {"name": "Suresh", "age": 30}
]
def sort_by_age(user_list):
    return sorted(user_list, key=lambda user: user["age"])

print(sort_by_age(users))
```

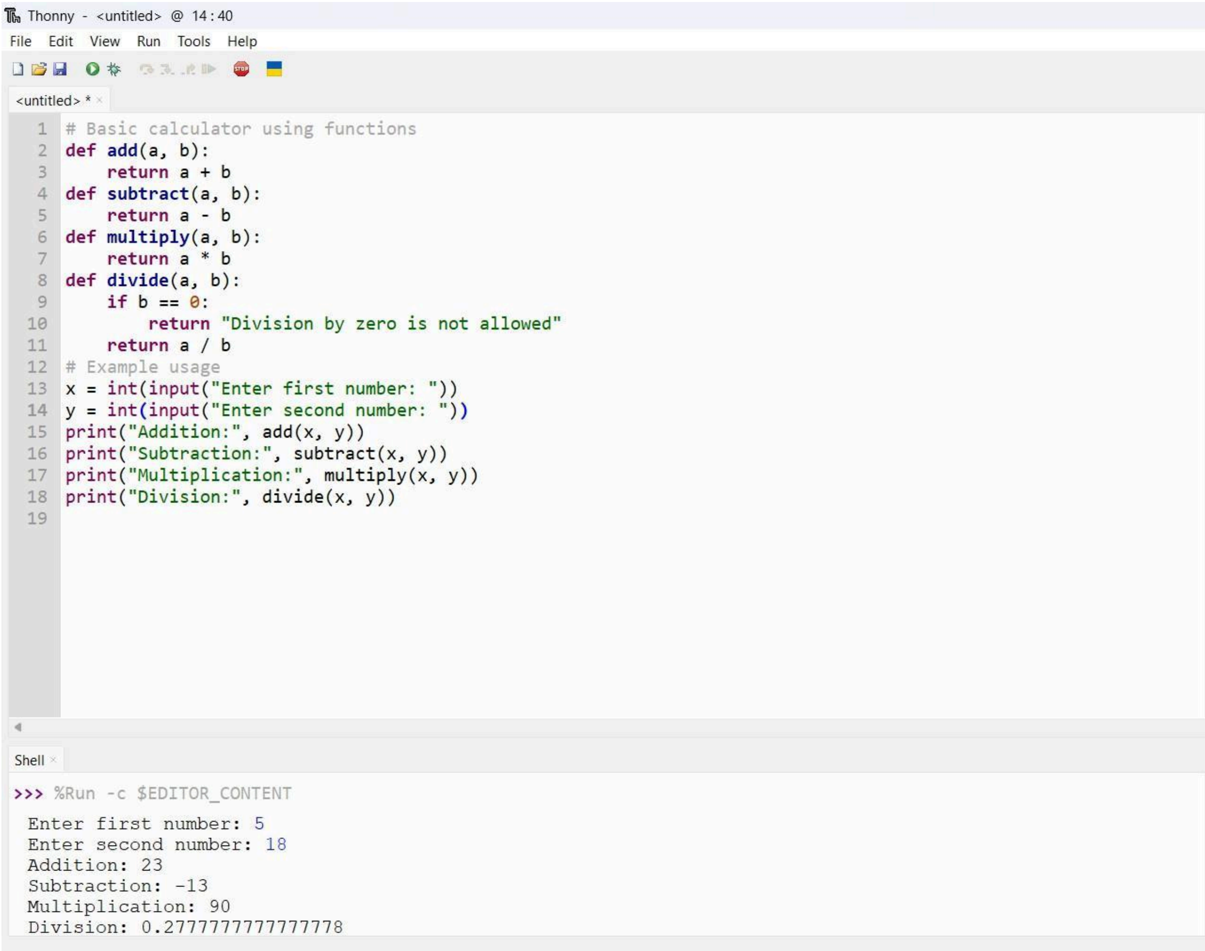
```
>>> %Run -c $EDITOR_CONTENT
 [{'name': 'Anita', 'age': 22}, {'name': 'Ravi', 'age': 25}, {'name':
 'Suresh', 'age': 30}]
>>>
```

Comparison (Short Note):

The solution generated by Gemini is short and direct, making it quick to understand for simple tasks. Cursor AI, on the other hand, focused more on clean structure by using a function, which improves readability and makes the code reusable. While both approaches have similar performance, the Cursor AI version is more suitable for larger projects where maintainability and clarity are important.

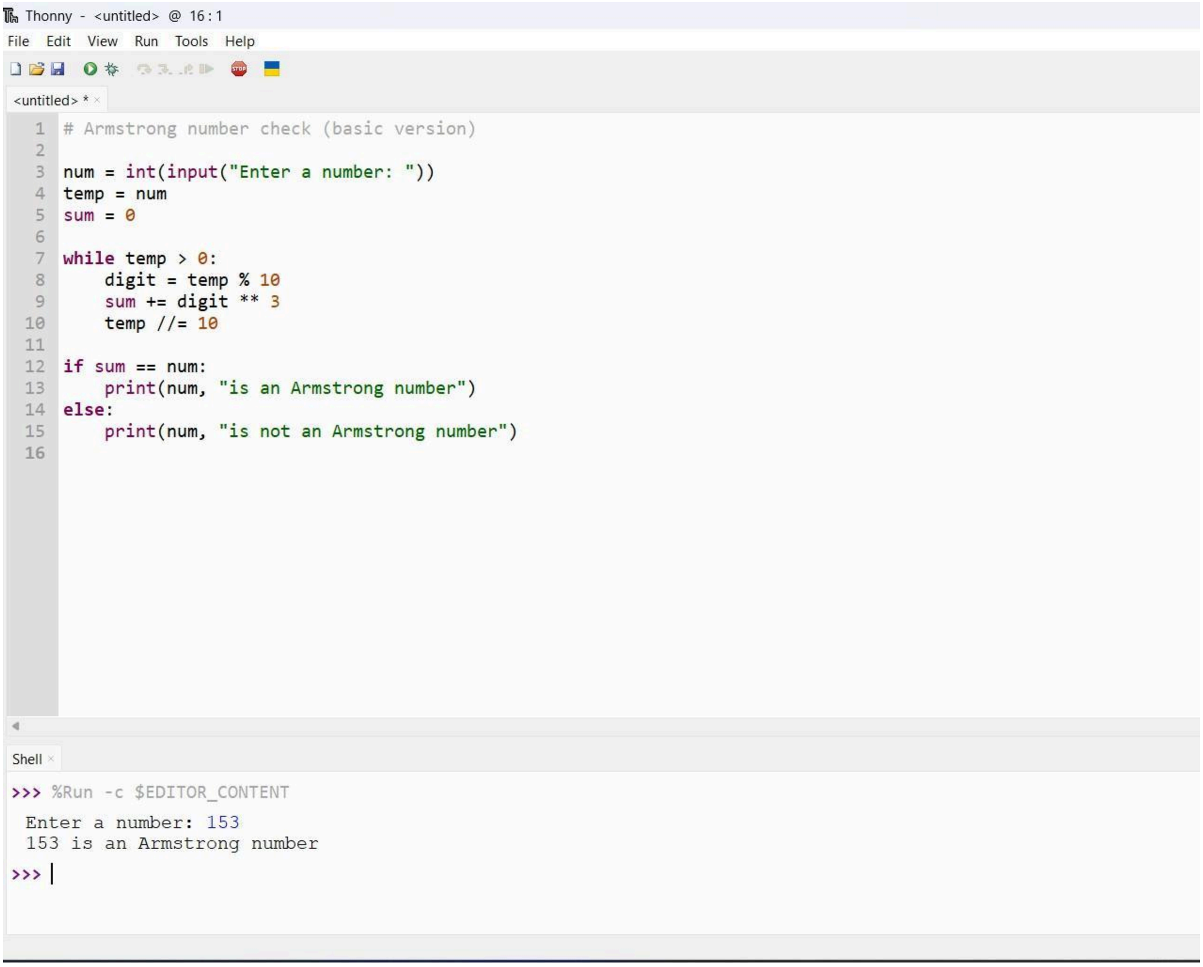
Task 3: Calculator Using Functions (Gemini)

Calculator Code

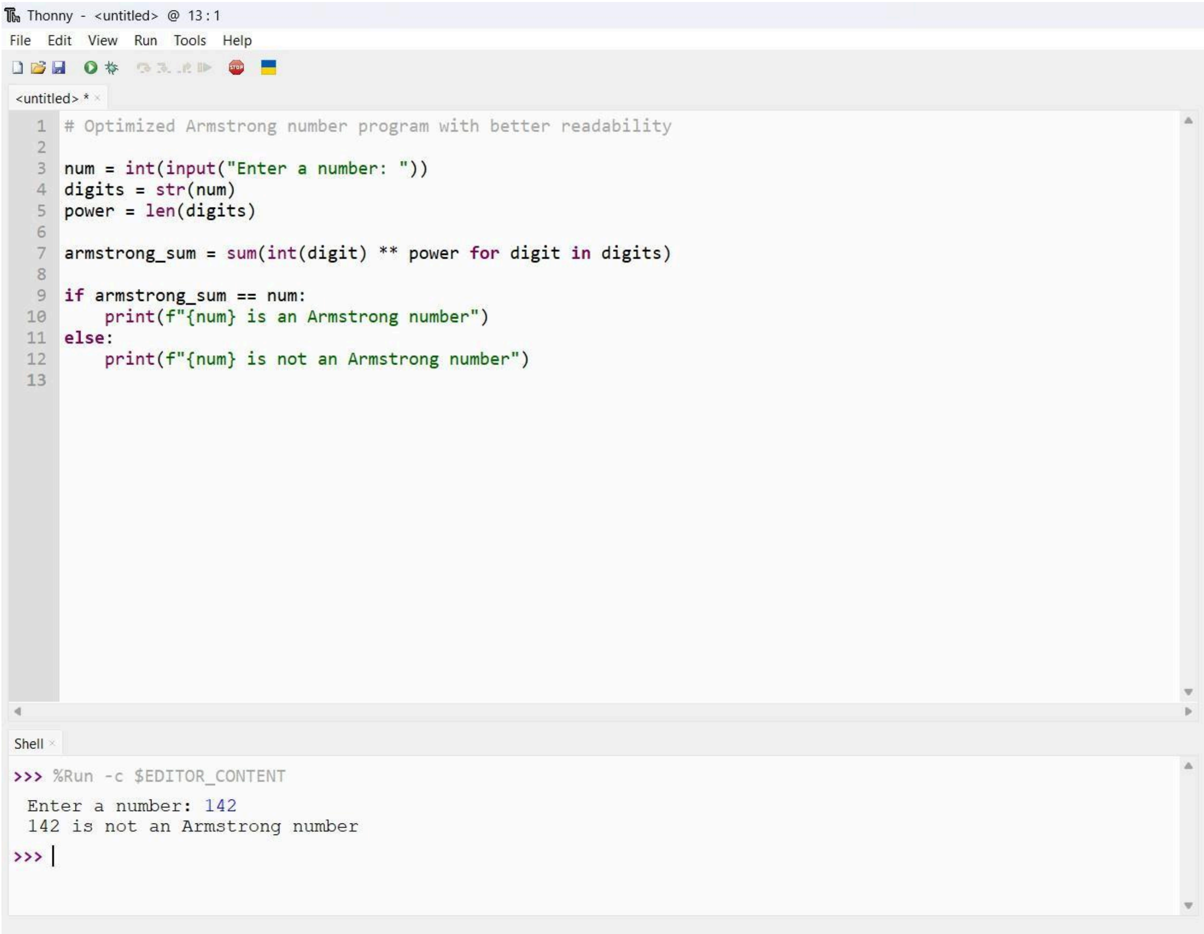


Task 4: Armstrong Number Optimization

◆ **Version 1: Gemini-Generated Armstrong Program**



Version 2: Optimized Using Cursor AI



Summary of Improvements:

- **Reduced lines of code**
- **Removed unnecessary variables**
- **Improved readability using Python built-in functions**
- **Works for Armstrong numbers of any length**